FALCO: an operational active-learning framework with progressive LLM oracles for long-tailed short-text classification — design, zero-cost execution, and honest validation

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Abstract

Surveys of practitioners report that the barriers to adopting active learning (AL) with large language models are operational — deployment complexity, uncertain cost reduction, inadequate tooling — rather than accuracy. FALCO answers with a framework whose every decision is an audited, pre-registered instrument: a label-free cold start (DRI-SL); uncertainty-driven selection with a cheap classifier in the loop; a *progressive oracle* policy where a budget-quality menu of LLM annotators is measured first (accuracy with confidence intervals, cost per thousand labels with cache and batch accounting) and a pre-registered gate decides which oracle serves each phase; stagnation-based stopping; and a release-plus-drift policy that reuses the cheap oracle as a continuous audit instrument. We report a complete end-to-end execution at *zero oracle cost* on a 621-class Portuguese retail task: with a free-tier LLM oracle (≈78–80% accuracy), the loop stopped itself by validation stagnation at 32–40% of a 15,000-label budget. Validation deliberately keeps the strong classifier (BERTimbau) out of the loop, training it once per arm on collected sets: oracle noise costs 4.4 accuracy points against gold labels on identical instances; actively selected sets cover 620 of 649 classes against 493 for random sets of equal size; and an entropy-selected 15k set reaches 83.1% population accuracy. The pre-registered hypothesis ($F1 \geq 0.95$ of full-pool supervision at $\leq 30\%$ labels) is *not* sustained at 30% but **holds from 50% of the pool** (a budget sweep locates the floors: 40% for accuracy, 50% for macro-F1); at 70% the actively selected set *beats* full-pool supervision on both metrics. The paired decomposition shows why the 30% budget fell short: the binding constraint is neither oracle noise nor selection but the stopping criterion inherited from the light classifier, which halts the loop long before the strong classifier stops converting labels into tail coverage. Noise-robustness experiments bound the damage of oracle error (87% of final macro-F1 retained at 10% uniform noise, 74% at 20%), and the framework's measurement layer contributed two instrument-level findings (output-contract artifact; serving-provider divergence) reported in companion work. Code, prompts, per-annotation artifacts and a reproducible runner are released.

Keywords: active learning; large language models; data annotation; annotation cost; text classification; MLOps.

1 Introduction

Two decades of active learning research minimized *label counts* under an assumed-correct oracle (Settles, 2012; Olsson, 2009); cost-aware work showed counts are not costs (Settles et al., 2008); and the LLM-as-annotator turn (Gilardi et al., 2023; Zhang and Takada, 2025; Rouzegar and Makrehchi, 2024; Kholodna et al., 2024) changed the currency twice: cost became a function of tokens (and caching, and batching), and oracle quality became a *purchasable parameter* on a menu spanning an order of magnitude in price. Yet community surveys (Romberg et al., 2025) find adoption blocked by the same barriers as fifteen years ago: deployment complexity, uncertain real savings, inadequate tooling.

FALCO (*Framework de Aprendizado Ativo com LLM para texto CurtO*) is our answer: an AL framework for long-tailed short-text classification whose distinguishing property is not a new selection strategy but an *operational discipline* — every component decision (which oracle, which phase, when to stop, when to release, when to retrain) is derived from a measured, pre-registered, artifact-backed criterion. This paper presents the framework and its complete execution and validation on a hard industrial task: 621 imbalanced product categories, Portuguese fiscal-receipt descriptions of 4–50 characters.

2 The framework

FALCO composes four components, each validated separately in companion work and integrated here. [TODO: figura da arquitetura (fases + gate + ciclo de vida) — obrigatória para ESWA]

Phase 1 — label-free cold start. The initial set is built by DRI-SL, a deterministic heuristic (semantic-density clustering plus lexical-variety filling) that outperforms even a supervised evolutionary optimizer of L_0 composition in the 100–5,000 range. Phase 1 spends at most 1% of the label budget.

Phase 2 — uncertainty selection with a cheap learner. The loop retrains a light classifier per batch (per-class prototype PVBin or logistic SGD (Darú, 2024)); entropy selection dominates at scale (saturating at 95% of its ceiling with 16–38% of the pool, versus 80%+ for random). The strong classifier stays *out* of the loop by design — it is trained once, at the end, on the collected set; this is both the cost model and, as Section 4 shows, the correct validation design.

Progressive oracle with a pre-registered gate. Candidate LLM oracles are measured first on paired samples (Wilson intervals; McNemar comparisons; cost per thousand labels with provider-metered tokens, cache and batch amortization). A gate fixed *before* measurement maps results to architecture: the Initial oracle is the best accuracy/cost ratio; an Advanced oracle exists only if statistically superior to the Initial ($\alpha = 0.05$); if no candidate clears an 85% accuracy bar, noisy-oracle training becomes the central scenario and robustness (below) is mandatory. On our task the gate selected deepseek-v4-flash (78–82% at US\$0.035/1k) as Initial and deepseek-v4-pro as Advanced ($p < 0.001$), with a zero-cost free-tier oracle (nemotron-3-ultra, statistically tied with the Initial, $p = 0.76$) as the no-budget alternative.

Stopping, release, drift. The loop stops on the first of: validation-macro-F1 stagnation ($< 10^{-3}$ gain over 5 iterations), budget exhaustion, or pool exhaustion. Release requires stagnation *plus* the validation Wilson interval clearing a floor; the untouched test set is consumed exactly once. In production, three monitoring layers (input drift at zero label cost; confidence drift; weekly oracle spot-audit at ≈US$0.01 per 200 items) trigger incremental re-runs of the loop over recent data.

3 Zero-cost end-to-end execution

The full loop ran twice (PVBin and SGD in the loop) with the free oracle and a 15,000-label budget over a 50,000-instance pool. Both runs **stopped themselves** by validation stagnation — at 6,009 labels (PVBin) and 4,742 (SGD), i.e., 32–40% of budget — with validation and test curves flat at the stop point, and 241 invalid oracle responses out of 6,250 oracle calls in the first run (3.9%) handled by the framework's strict validation layer. Monetary oracle cost: zero; the binding constraint was provider throughput, which the framework's batch calibration (paired McNemar across batch sizes) and caching layer are designed to manage.

4 Validation with the strong classifier out of the loop

BERTimbau (Souza et al., 2020) is fine-tuned once per arm on already-collected sets and evaluated on a reserved population (≈177k instances) never touched by selection, with a fixed stratified evaluation sample and Wilson intervals. Arms: (A) the 8,937 items annotated by the real pipeline, with the *oracle's* labels; (B) the same items with gold labels; (C) equal-size random sets with gold; (D) the full 50k pool with gold — the ruler; (E) a 15k entropy-selected prefix with gold.

Findings so far: oracle noise costs **4.4 accuracy points** (A: 65.9% [65.2, 66.5] vs. B: 70.2% [69.6, 70.9]) — the price of 71.6% oracle-gold agreement on *actively selected* (hence harder) instances; random selection wins raw accuracy at equal size (C: 73.2%) while active selection wins class coverage (620 vs. 493 classes) and macro-F1 (A: 0.242, B: 0.209, C: 0.189); the full pool (D) sets the ruler at 88.3% accuracy / 0.451 macro-F1.

The budget floor that sustains the hypothesis. A budget sweep (BERTimbau on entropy-selected prefixes of 20/25/30/35k gold labels, Table 1) locates where the strong classifier crosses 0.95× the ruler. The pre-registered hypothesis (≥ 0.95 of full-pool macro-F1 at $\leq 30\%$ labels) is refuted at 30% but **holds from 50% of the pool**: accuracy crosses at 40% (20k), macro-F1 at 50% (25k). Crucially, 70% of actively selected labels (35k) *beat* the full pool on both metrics (88.6% vs. 88.3%; 0.463 vs. 0.451) — labeling everything is counterproductive, the "less is more" effect reappearing on the transformer. The bottleneck was the stopping criterion, not the oracle: the framework sustains the hypothesis with a strong-model-anchored stop or a ≈50% budget floor.

Robustness to oracle noise. Controlled uniform-noise experiments with the light classifier bound the damage: 87% of final macro-F1 retained at $\varepsilon = 0.1$, 74% at $\varepsilon = 0.2$, 54% at $\varepsilon = 0.4$. Since real oracle error is *structured* (concentrated in sibling categories), and structured noise of equal rate is more benign than uniform (Frénay and Verleysen, 2014; Song et al., 2023) — with the caveat that real-noise benchmarks warn against overclaiming (Merdjanovska et al., 2024; Rączkowska et al., 2024) — the uniform design is a deliberately pessimistic test.

Table 1: Budget sweep: BERTimbau on entropy-selected prefixes, reserved population. Criterion: acc $\geq$ 83.9%, macro-F1 $\geq$ 0.428.

Arm	**Labels**	**% pool**	**Acc.**	**Macro-F1**
A	8,937	18%	65.9%	0.242
E	15,000	30%	83.1%	0.380
E20	20,000	40%	85.9%	0.418
E25	25,000	50%	87.3%	**0.434**
E30	30,000	60%	87.7%	0.456
E35	35,000	70%	**88.6%**	**0.463**
D	50,000	100%	88.3%	0.451

5 Related work

LLM-in-the-loop AL spans annotator (Zhang and Takada, 2025), selector (Bayer and Reuter, 2024), confidence routing (Rouzegar and Makrehchi, 2024), batch annotation (Kholodna et al., 2024), candidate-label distillation (Xia et al., 2025), LLM mixtures (Qi et al., 2026) and dual-expert taxonomy contraction (Cheng et al., 2024); cold-start seeding is addressed by DEUCE (Guo et al., 2025) among others. Applied Portuguese retail classification at scale exists without AL or oracle automation (Machado and Veloso, 2026). FALCO's contribution is not any single mechanism but the integration under measurement discipline — the combination of informed cold start + progressive measured oracle + instrumented cost + pre-registered decisions, which, to our knowledge, no prior framework offers end to end.

6 Conclusion

FALCO shows that the practical barriers to LLM-assisted active learning — cost uncertainty, deployment opacity, unverifiable claims — are addressable by treating measurement as part of the framework: oracles are purchased off a measured menu through a pre-registered gate; the loop stops itself; the strong model trains once; release requires a reserved-set criterion; and production monitoring reuses the cheap oracle as an audit instrument. A complete execution at zero oracle cost, stopping at a third of budget with the strong-classifier validation quantifying every loss source (noise vs. selection vs. budget), demonstrates the discipline is affordable. The pre-registered criterion returned a negative verdict with an identified, correctable cause — the outcome pre-registration exists to protect: the framework's measurement layer converted a failed hypothesis into a design correction, which is precisely what an operational framework owes its adopters.

Artifacts. `github.com/GHDaru/activelearning` (runner, prompts, per-annotation cache, experiment states); `github.com/GHDaru/tesedaru` (pre-registered criteria).

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